Fiscal Year:	FY 2024	**Task Last Updated:**	FY 03/27/2024
PI Name:	Wyatt, Sarah E Ph.D.		
Project Title:	Step 2: Comparative Analysis of Multi-Gravity Studies on Earth and ISS		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2020 Space Biology NNH20ZDA001N-SB E.12. Flight/Ground Research
Start Date:	12/15/2021	**End Date:**	05/30/2023
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA KSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 05/30/2023 per NSSC information (Ed, 4/18/23)		
Key Personnel Changes/Previous PI:	The postdoctoral scholar originally associated with this grant moved to a NASA postdoc. Thus, although the study is on course with the personnel we have, an extended timeframe is needed to complete the analysis.		
COI Name (Institution):	Meyers, Alexander Ph.D. (Ohio University)		
Grant/Contract No.:	80NSSC22K0366		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	In response to the ROSES2020 Space Biology solicitation for NASA GeneLab Analytical Investigations, we propose a comparative meta-analysis of available fractional-g datasets in Arabidopsis to address Focus Area B: combined effects of space-relevant stressors and altered gravity. Over the past 20 years, the space biology community has made great advances in understanding plant transcriptional networks under Earth's 1g and the microgravity environment of spaceflight. However, beyond-Earth exploration will require plant growth at Lunar (0.16g) and Martian (0.38g) gravities. Molecular data at these fractional g levels is sparse and offers little information on the interactions between partial gravity signaling networks and other stressors. We propose to use an integrative meta-analysis approach combining comparative differential expression, contrast analyses, analyses of the plant responses across the gravitational continuum, and weighted gene co-expression network analyses (WGCNA) of the five available partial-g, transcriptomic datasets, to: (Aim 1) identify core trends in molecular/physiological responses to partial g, (Aim 2) leverage the diversity of the experimental factors in those experiments to delineate the influences of those influences, and (Aim 3) use the series of 1g flight and ground controls across the datasets to identify specific artifacts of simulated micro and partial g. Three of the datasets (Seedling Growth 1, 2, and 3) were flown using European Modular Cultivation System (EMCS) experiments that used variation of 4 genotypes, and 5 different g levels under unidirectional blue or red light aboard the International Space Station (ISS). The second dataset comes from the EMCS-Plant Gravity Perception (PGP) spaceflight experiment, which resulted in tissue-specific transcriptomes at 12g levels between 0.003 and 1g. And the third derives from a series of experiments using simulated partial gravity on ground analogs to treat seedlings at simulated 0g, 0.38g, 1g, and 2g. By leveraging all of these datasets, we can determine molecular and physiological changes that result from gravitational intensity and other environmental stressors. The project directly addresses several questions relating to plant growth and development and the functional use of existing datasets to answer novel questions; i.e., PB-1 ("How does gravity affect plant growth, development & metabolism?" and PB-5 ("How do plants sense and react to gravity and what are the molecular mechanisms involved?"), GL-2 ("Can raw datasets, including unprocessed molecular data that can be subjected to subsequent analyses lead to increased use of ISS data to facilitate new science products?") GL-3 ("What are the key cellular and molecular systems necessary for biological organisms to thrive in the space environment that can be discovered from 21st century bioinformatics tools?")
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Potential Impact: This project will contribute to our understanding of plant growth in partial-g as will be needed for beyond-earth plant cultivation and human space exploration/colonization. This study has been successful in identifying key differences between true and simulated gravity which might increase the accuracy of on Earth simulations in preparation for life beyond Earth. Additionally, because this study was conducted using only open data, it has become an example of the successes that are possible when data is made open and FAIR. The presentations of this data have encouraged others to release open data which has resulted in additional data being added to the GeneLab data repository and multiple past PIs updating their metadata to increase accessibility. Examples of successful analyses of open data are vital when advocating for more researchers to make their primary data available. _____
Task Progress:	Here, we used an integrative meta-analyses approach to identify genes impacting gravitropic signaling. We used datasets housed on the NASA GeneLab data repository. Five RNA sequencing (RNA-seq) datasets have been produced from five different experiments from three different research laboratories. All used Arabidopsis seedlings which had been exposed to real or simulated fractional gravity. Of these datasets, four were conducted on the International Space Station (ISS) using the European Modular Cultivation System (EMCS). A fifth experiment was conducted using the facilities at the European Space Research and Technology Center (ESTEC), using random positioning machines (RPMs) to simulate microgravity and RPM centrifuges to simulate fractional gravity and 1g, as well as a large diameter centrifuge to simulate hyper gravity. Combined, these data represent two ecotypes of Arabidopsis, three mutants, four plant organs, and thirteen gravity levels. These data contain over 150 distinct experimental conditions. The variability in design between experiments allows us to not only identify consistently differential expressed genes in spaceflight, but also compare simulated fractional gravity on Earth with fractional gravity in space. Together, these analyses provide a better understanding of the unique nature of simulated gravity as compared to spaceflight and ground controls. Three computation approaches were used for the analysis of these five datasets. The comparison of: 1) Differentially expressed genes showed conserved gene expression across multiple flights. 2) Weighted Gene Co-expression network analysis identified groups of interacting genes. 3) Sequential analysis using ImpulseDE2 was adapted to work with increasing gravity intensities. We were successful in finding conserved responses to micro, lunar, and Martian gravity. This success was seen in the results from ImpulseDE2. This analysis tool is usually used to find changes in gene expression of sequential time points. In this meta-analysis, it was used to find expression trends in sequential gravity intensities. ImpulseDE2 found three transcriptional groupings on the gravity gradient. These groupings of genes are expressed at low levels of gravity, <0.18g, Earth like gravity, >0.65g, and the intermediate gravity levels between those points, 0.18 < intermediate g <0.65g. We found this trend was conserved across all spaceflight data sets, including both ecotypes of Arabidopsis and all three mutant lines. From the ImpulseDE2 results, we clearly see core trends in transcriptomic regulation not only in microgravity and Earth gravity, but at important intermediate gravity levels. To evaluate the utility of partial gravity studies, the differential expression for the PGP-ESTEC dataset was compared to differential expression from spaceflight datasets. The PGP Flight, PGP-ESTEC, and SG2 datasets all include the Columbia ecotype, while SG1 and SG3 do not. To compare the effects of different types of gravity stimulation (true or simulated), the Columbia plants from these three datasets were compared. Microgravity and Martian gravity were selected to avoid spaceflight being a variable, as the SG2 dataset kept its 1g control on Earth. Data normalization and differential expression was calculated separately for each of the datasets and results were compared. The spaceflight datasets consistently showed more similarity to other spaceflight data than they did to simulated gravity conditions. When quantifying the overlap in differentially expressed genes (DEGs) for microgravity to Martian gravity, the PGP-ESTEC transcriptome showed the lowest number of shared genes. At first, this difference might appear to be caused by fewer total differentially expressed genes identified in the PGP-ESTEC dataset. However, this does not explain why only 63% of the DEGs in the ESTEC data were shared, with other experiments while the other datasets averaged 81% gene overlap. These findings suggest that simulated gravity does not induce an equivalent response to true gravity.

	From the differential expression, we could see the effects of spaceflight outside of gravity had a substantial impact on what genes were shared between datasets. The PGP Flight experiment grew plants under microgravity and 1g on the ISS. Seedling Growth 2 (SG2) grew plants in microgravity on the ISS but kept their 1g plants as a ground control on Earth. PGP-EMCS simulated 1g and microgravity on Earth using random positioning machines (RPMs) and RPM centrifuges. Because spaceflight is a variable between these datasets, but gravity is not, we can identify the effects of other spaceflight stressors while controlling for the effects of gravity. The impact of radiation, fluid effects, and other spaceflight variables was quantified by the lack of overlap between datasets at 1g. The large overlap within experiment, compared to the small overlap between experiments, highlights the importance of other spaceflight variables. Each additional stress to the plant produces a compounding stress that can only be induced when all stresses of spaceflight are present. To further identify which gene families were regulated in response to the stress of spaceflight, weighted gene co-expression was used to find gene clusters that shifted together between datasets. While this analysis did provide a list of potentially interesting genes, the main differences highlighted were across genotypes and RNA extraction types, as opposed to environmental change. Although these results are not immediately helpful in interpreting transcriptome responses to spaceflight, it does give insights into data approaches for open science. Not all these datasets used long reads, paired reads, or synthetic RNA spike-ins. Because of this inherent difference, general normalization techniques had to be used across fundamentally different extraction types for the weighted gene contrast analysis. These findings highlight the importance of setting technical standards for biological extractions on spaceflight tissues. The point in the extractions where spike-ins are added can also be a source of technical variation. The variety of experimental designs used by primary investigators has provided a wealth of data for this meta-analysis. The differences in genotype, controls, and lighting provide evidence of conserved response pathways to the only consistent feature, gravity. While design differences have strengthened the analysis, consistency in technical approaches to RNA sequencing could add to the accessibility of future open data.
Bibliography Type:	Description: (Last Updated: 11/25/2025)
Abstracts for Journals and Proceedings	Canaday E, Wyatt SE. "Changes in Moon, Mars, and micro gravity: A meta-analysis of transcriptome shifts in response to gravity and spaceflight." 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. Abstracts. 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. , Nov-2023
Abstracts for Journals and Proceedings	Canaday E, Wyatt SE. "Changes in Moon, Mars, and micro gravity: A meta-analysis of transcriptome shifts in response to gravity and spaceflight." Plant Biology 2023, Savannah, GA, August 5-9, 2023. Abstracts. Plant Biology 2023, Savannah, GA, August 5-9, 2023. , Aug-2023
Abstracts for Journals and Proceedings	Wyatt SE. "Plant space biology: What we can learn from RNA sequencing." NASA GeneLab for High Schools (GL4HS), Moffett Field, California, June 13-July 8, 2022. Abstracts. GeneLab for High Schools (GL4HS), Moffett Field, California, June 13-July 8, 2022. , Jun-2022
Articles in Peer-reviewed Journals	Barker R, Kruse CP, Johnson C, Saravia-Butler A, Fogle H, Chang HS, Trane RM, Kinscherf N, Villacampa A, Manzano A, Herranz R, Davin LB, Lewis NG, Perera I, Wolverton C, Gupta P, Reinsch SS, Wyatt SE, Gilroy S. "Meta-analysis of the space flight and microgravity response of the Arabidopsis plant transcriptome." npj Microgravity. 2023 Mar 20;9(1):21. https://doi.org/10.1038/s41526-023-00247-6 ; PMID: 36941263; PMCID: PMC10027818 , Mar-2023
Articles in Peer-reviewed Journals	Meyers A, Wyatt SE. "Plant space biology in the genomics age." Annual Plant Reviews. Online. 2022 May 4;5(2). https://doi.org/10.1002/9781119312994.apr0784 ; PMID: 19694953 , May-2022